MATH329 · OPERATIONS RESEARCH

Big-M Practice Solutions

6 problems: from the slides, quiz 471, and practice

Muharib

01 Solution 1: min

1 Standard form + R

Both constraints are `≥`, so each takes `−s` and `+R`. In min, every R gets `+M` in the objective.

$$\begin{aligned} &\text{Minimize } z = 2x_1 + x_2 + 0s_1 + 0s_2 + MR_1 + MR_2 \\ &\text{subject to} \quad 3x_1 + x_2 - s_1 + R_1 = 9 \\ &\qquad\qquad\quad x_1 + x_2 - s_2 + R_2 = 6 \\ &\qquad\qquad\quad x_1,\ x_2,\ s_1,\ s_2,\ R_1,\ R_2 \geq 0 \end{aligned}$$

2 The original Z row

Move the whole objective to one side, so every coefficient flips sign:

	x_1	x_2	s_1	s_2	R_1	R_2	Sol.
Z	−2	−1	0	0	−M	−M	0

The problem: R_1 and R_2 are basic, but its Z-row entry is not zero. It must become zero.

3 Fixing the Z row

It is a min problem, so multiply every R row by M and add it to the Z row. Add each column from top to bottom:

	x_1	x_2	s_1	s_2	R_1	R_2	Sol.
Old Z	−2	−1	0	0	−M	−M	0
+ M × R_1	3M	M	−M	0	M	0	9M
+ M × R_2	M	M	0	−M	0	M	6M
New Z	4M − 2	2M − 1	−M	−M	0	0	15M

Check: every R column is now 0 ✓

4 The first tableau

Basic	x_1	x_2	s_1	s_2	R_1	R_2	Sol.
R_1	3	1	−1	0	1	0	9
R_2	1	1	0	−1	0	1	6
Z	4M − 2	2M − 1	−M	−M	0	0	15M

5 Iteration 1

1. Entering: We look for the most positive entry in the Z row. Compare the coefficient of M first.

x_1: $4M - 2$ x_2: $2M - 1$

Entering: x_1

2. Leaving: Divide Sol by the entering column, positive entries only. The smallest ratio leaves.

Basic	x_1	Sol.	Ratio
R_1	3	9	3
R_2	1	6	6

Leaving: R_1 · The pivot = 3

3. The new pivot row

$$\text{New } x_1 = R_1 \div 3 = \left(1, \frac{1}{3}, -\frac{1}{3}, 0, \frac{1}{3}, 0 \mid 3\right)$$

4. The other rows: Each row = old row − (its entry in the entering column) × the new pivot row.

$$\text{New } R_2 = R_2 - (1) \times \text{New } x_1 = \left(0, \frac{2}{3}, \frac{1}{3}, -1, -\frac{1}{3}, 1 \mid 3\right)$$

$$\text{New } Z = Z - (4M - 2) \times \text{New } x_1 = \left(0, \frac{2M}{3} - \frac{1}{3}, \frac{M}{3} - \frac{2}{3}, -M, \frac{2}{3} - \frac{4M}{3}, 0 \mid 3M + 6\right)$$

5. The new tableau

Basic	x_1	x_2	s_1	s_2	R_1	R_2	Sol.
x_1	1	$\frac{1}{3}$	$-\frac{1}{3}$	0	$\frac{1}{3}$	0	3
R_2	0	$\frac{2}{3}$	$\frac{1}{3}$	−1	$-\frac{1}{3}$	1	3
Z	0	$\frac{2M}{3} - \frac{1}{3}$	$\frac{M}{3} - \frac{2}{3}$	−M	$\frac{2}{3} - \frac{4M}{3}$	0	3M + 6

6 Iteration 2

1. Entering: We look for the most positive entry in the Z row. Compare the coefficient of M first.

x_2: $\frac{2M}{3} - \frac{1}{3}$ $\quad$ s_1: $\frac{M}{3} - \frac{2}{3}$

Entering: x_2

2. Leaving: Divide Sol by the entering column, positive entries only. The smallest ratio leaves.

Basic	x_2	Sol.	Ratio
x_1	$\frac{1}{3}$	3	9
R_2	$\frac{2}{3}$	3	$\frac{9}{2}$

Leaving: R_2 · The pivot = $\frac{2}{3}$

3. The new pivot row

$$\text{New } x_2 = R_2 \div \frac{2}{3} = (0,\ 1,\ \frac{1}{2},\ -\frac{3}{2},\ -\frac{1}{2},\ \frac{3}{2} \mid \frac{9}{2})$$

4. The other rows: Each row = old row − (its entry in the entering column) × the new pivot row.

$$\text{New } x_1 = x_1 - (\frac{1}{3}) \times \text{New } x_2 = (1,\ 0,\ -\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2},\ -\frac{1}{2} \mid \frac{3}{2})$$

$$\text{New } Z = Z - (\frac{2M}{3} - \frac{1}{3}) \times \text{New } x_2 = (0,\ 0,\ -\frac{1}{2},\ -\frac{1}{2},\ \frac{1}{2} - M,\ \frac{1}{2} - M \mid \frac{15}{2})$$

5. The new tableau

Basic	x_1	x_2	s_1	s_2	R_1	R_2	Sol.
x_1	1	0	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{3}{2}$
x_2	0	1	$\frac{1}{2}$	$-\frac{3}{2}$	$-\frac{1}{2}$	$\frac{3}{2}$	$\frac{9}{2}$
Z	0	0	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2} - M$	$\frac{1}{2} - M$	$\frac{15}{2}$

7 The answer

No positive entry is left in the Z row (M is large), so the tableau is optimal.

$$\text{Min } z = \frac{15}{2} \quad \text{at} \quad x_1 = \frac{3}{2}, \quad x_2 = \frac{9}{2}$$

02 Solution 2: max with an = constraint

1 Standard form + R

The `=` constraint takes only `R1`, and the `≥` constraint takes `−s2` and `R2`. In max, every R gets `−M`.

$$\begin{aligned} &\text{Maximize } z = 2x_1 + 3x_2 - 5x_3 + 0s_2 - MR_1 - MR_2 \\ &\text{subject to} \quad x_1 + x_2 + x_3 + R_1 = 7 \\ &\qquad\qquad\quad 2x_1 - 5x_2 + x_3 - s_2 + R_2 = 10 \\ &\qquad\qquad\quad x_1, x_2, x_3, s_2, R_1, R_2 \geq 0 \end{aligned}$$

2 The original Z row

Move the whole objective to one side, so every coefficient flips sign:

	x_1	x_2	x_3	s_2	R_1	R_2	Sol.
Z	−2	−3	5	0	M	M	0

The problem: R_1 and R_2 are basic, but its Z-row entry is not zero. It must become zero.

3 Fixing the Z row

It is a max problem, so multiply every R row by M and subtract it from the Z row. Work each column from top to bottom:

	x_1	x_2	x_3	s_2	R_1	R_2	Sol.
Old Z	−2	−3	5	0	M	M	0
− M × R_1	−M	−M	−M	0	−M	0	−7M
− M × R_2	−2M	5M	−M	M	0	−M	−10M
New Z	−3M − 2	4M − 3	5 − 2M	M	0	0	−17M

Check: every R column is now 0 ✓

4 The first tableau

Basic	x_1	x_2	x_3	s_2	R_1	R_2	Sol.
R_1	1	1	1	0	1	0	7
R_2	2	−5	1	−1	0	1	10
Z	−3M − 2	4M − 3	5 − 2M	M	0	0	−17M

5 Iteration 1

1. Entering: We look for the most negative entry in the Z row. Compare the coefficient of M first.

x_1: $-3M - 2$ x_3: $5 - 2M$

Entering: x_1

2. Leaving: Divide Sol by the entering column, positive entries only. The smallest ratio leaves.

Basic	x_1	Sol.	Ratio
R_1	1	7	7
R_2	2	10	5

Leaving: R_2 · The pivot = 2

3. The new pivot row

$$\text{New } x_1 = R_2 \div 2 = \left(1, -\frac{5}{2}, \frac{1}{2}, -\frac{1}{2}, 0, \frac{1}{2} \mid 5\right)$$

4. The other rows: Each row = old row − (its entry in the entering column) × the new pivot row.

$$\text{New } R_1 = R_1 - (1) \times \text{New } x_1 = \left(0, \frac{7}{2}, \frac{1}{2}, \frac{1}{2}, 1, -\frac{1}{2} \mid 2\right)$$

$$\text{New } Z = Z - (-3M - 2) \times \text{New } x_1 = \left(0, -\frac{7M}{2} - 8, 6 - \frac{M}{2}, -\frac{M}{2} - 1, 0, \frac{3M}{2} + 1 \mid 10 - 2M\right)$$

5. The new tableau

Basic	x_1	x_2	x_3	s_2	R_1	R_2	Sol.
R_1	0	$\frac{7}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$-\frac{1}{2}$	2
x_1	1	$-\frac{5}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	5
Z	0	$-\frac{7M}{2} - 8$	$6 - \frac{M}{2}$	$-\frac{M}{2} - 1$	0	$\frac{3M}{2} + 1$	$10 - 2M$

6 Iteration 2

1. Entering: We look for the most negative entry in the Z row. Compare the coefficient of M first.

x_2: $-\frac{7M}{2} - 8$ x_3: $6 - \frac{M}{2}$ s_2: $-\frac{M}{2} - 1$

Entering: x_2

2. Leaving: Divide Sol by the entering column, positive entries only. The smallest ratio leaves.

Basic	x_2	Sol.	Ratio
R_1	$\frac{7}{2}$	2	$\frac{4}{7}$
x_1	$-\frac{5}{2}$	5	— (not positive)

Leaving: R_1 · The pivot = $\frac{7}{2}$

3. The new pivot row

$$\text{New } x_2 = R_1 \div \frac{7}{2} = (0,\ 1,\ \frac{1}{7},\ \frac{1}{7},\ \frac{2}{7},\ -\frac{1}{7} \mid \frac{4}{7})$$

4. The other rows: Each row = old row − (its entry in the entering column) × the new pivot row.

$$\text{New } x_1 = x_1 - (-\frac{5}{2}) \times \text{New } x_2 = (1,\ 0,\ \frac{6}{7},\ -\frac{1}{7},\ \frac{5}{7},\ \frac{1}{7} \mid \frac{45}{7})$$

$$\text{New } Z = Z - (-\frac{7M}{2} - 8) \times \text{New } x_2 = (0,\ 0,\ \frac{50}{7},\ \frac{1}{7},\ M + \frac{16}{7},\ M - \frac{1}{7} \mid \frac{102}{7})$$

5. The new tableau

Basic	x_1	x_2	x_3	s_2	R_1	R_2	Sol.
x_2	0	1	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{2}{7}$	$-\frac{1}{7}$	$\frac{4}{7}$
x_1	1	0	$\frac{6}{7}$	$-\frac{1}{7}$	$\frac{5}{7}$	$\frac{1}{7}$	$\frac{45}{7}$
Z	0	0	$\frac{50}{7}$	$\frac{1}{7}$	$M + \frac{16}{7}$	$M - \frac{1}{7}$	$\frac{102}{7}$

7 The answer

No negative entry is left in the Z row (M is large), so the tableau is optimal.

$$\text{Max } z = \frac{102}{7} \quad \text{at} \quad x_1 = \frac{45}{7}, \quad x_2 = \frac{4}{7}, \quad x_3 = 0$$

03 Solution 3: quiz 471

1 Standard form + R

The $\leq$ constraint takes only s_1 (basic from the start), and the $\geq$ constraint takes $-s_2$ and R. In max, R gets $-M$.

$$\begin{aligned} &\text{Maximize } z = 2x_1 + 3x_2 - 5x_3 + 0s_1 + 0s_2 - MR \\ &\text{subject to} \quad 4x_1 + 3x_2 - x_3 + s_1 = 24 \\ &\qquad\qquad\quad x_1 + x_2 + 2x_3 - s_2 + R = 18 \\ &\qquad\qquad\quad x_1, x_2, x_3, s_1, s_2, R \geq 0 \end{aligned}$$

2 The original Z row

Move the whole objective to one side, so every coefficient flips sign:

	x_1	x_2	x_3	s_1	s_2	R	Sol.
Z	−2	−3	5	0	0	M	0

The problem: `R` is basic, but its Z-row entry is not zero. It must become zero.

3 Fixing the Z row

It is a max problem, so multiply every R row by M and subtract it from the Z row.

Work each column from top to bottom:

	x_1	x_2	x_3	s_1	s_2	R	Sol.
Old Z	−2	−3	5	0	0	M	0
− M × R	−M	−M	−2M	0	M	−M	−18M
New Z	−M − 2	−M − 3	5 − 2M	0	M	0	−18M

Check: every R column is now 0 ✓

4 The first tableau

Basic	x_1	x_2	x_3	s_1	s_2	R	Sol.
s_1	4	3	−1	1	0	0	24
R	1	1	2	0	−1	1	18
Z	−M − 2	−M − 3	5 − 2M	0	M	0	−18M

5 Iteration 1

1. Entering: We look for the most negative entry in the Z row. Compare the coefficient of M first.

x_1: $-M - 2$ x_2: $-M - 3$ x_3: $5 - 2M$

Entering: x_3

2. Leaving: Divide Sol by the entering column, positive entries only. The smallest ratio leaves.

Basic	x_3	Sol.	Ratio
s_1	-1	24	— (not positive)
R	2	18	9

Leaving: R · The pivot = 2

3. The new pivot row

$$\text{New } x_3 = R \div 2 = \left(\frac{1}{2}, \frac{1}{2}, 1, 0, -\frac{1}{2}, \frac{1}{2} \mid 9\right)$$

4. The other rows: Each row = old row − (its entry in the entering column) × the new pivot row.

$$\text{New } s_1 = s_1 - (-1) \times \text{New } x_3 = \left(\frac{9}{2}, \frac{7}{2}, 0, 1, -\frac{1}{2}, \frac{1}{2} \mid 33\right)$$

$$\text{New } Z = Z - (5 - 2M) \times \text{New } x_3 = \left(-\frac{9}{2}, -\frac{11}{2}, 0, 0, \frac{5}{2}, M - \frac{5}{2} \mid -45\right)$$

5. The new tableau

Basic	x_1	x_2	x_3	s_1	s_2	R	Sol.
s_1	$\frac{9}{2}$	$\frac{7}{2}$	0	1	$-\frac{1}{2}$	$\frac{1}{2}$	33
x_3	$\frac{1}{2}$	$\frac{1}{2}$	1	0	$-\frac{1}{2}$	$\frac{1}{2}$	9
Z	$-\frac{9}{2}$	$-\frac{11}{2}$	0	0	$\frac{5}{2}$	$M - \frac{5}{2}$	−45

6 Iteration 2

1. Entering: We look for the most negative entry in the Z row.

$x_1: -\frac{9}{2}$ $x_2: -\frac{11}{2}$

Entering: x_2

2. Leaving: Divide Sol by the entering column, positive entries only. The smallest ratio leaves.

Basic	x_2	Sol.	Ratio
s_1	$\frac{7}{2}$	33	$\frac{66}{7}$
x_3	$\frac{1}{2}$	9	18

Leaving: s_1 · The pivot = $\frac{7}{2}$

3. The new pivot row

$$\text{New } x_2 = s_1 \div \frac{7}{2} = \left(\frac{9}{7}, 1, 0, \frac{2}{7}, -\frac{1}{7}, \frac{1}{7} \mid \frac{66}{7}\right)$$

4. The other rows: Each row = old row − (its entry in the entering column) × the new pivot row.

$$\text{New } x_3 = x_3 - \left(\frac{1}{2}\right) \times \text{New } x_2 = \left(-\frac{1}{7}, 0, 1, -\frac{1}{7}, -\frac{3}{7}, \frac{3}{7} \mid \frac{30}{7}\right)$$

$$\text{New } Z = Z - \left(-\frac{11}{2}\right) \times \text{New } x_2 = \left(\frac{18}{7}, 0, 0, \frac{11}{7}, \frac{12}{7}, M - \frac{12}{7} \mid \frac{48}{7}\right)$$

5. The new tableau

Basic	x_1	x_2	x_3	s_1	s_2	R	Sol.
x_2	$\frac{9}{7}$	1	0	$\frac{2}{7}$	$-\frac{1}{7}$	$\frac{1}{7}$	$\frac{66}{7}$
x_3	$-\frac{1}{7}$	0	1	$-\frac{1}{7}$	$-\frac{3}{7}$	$\frac{3}{7}$	$\frac{30}{7}$
Z	$\frac{18}{7}$	0	0	$\frac{11}{7}$	$\frac{12}{7}$	$M - \frac{12}{7}$	$\frac{48}{7}$

7 The answer

No negative entry is left in the Z row (M is large), so the tableau is optimal.

$$\text{Max } z = \frac{48}{7} \quad \text{at} \quad x_1 = 0\,, \quad x_2 = \frac{66}{7}\,, \quad x_3 = \frac{30}{7}$$

04 Solution 4: min with =, ≥ and ≤

1 Standard form + R

`=` takes R_1, `≥` takes $-s_2$ and R_2, and `≤` takes only s_3. In min, the R's get `+M`.

$$\begin{aligned}
&\text{Minimize } z = 4x_1 + x_2 + 0s_2 + 0s_3 + MR_1 + MR_2 \\
&\text{subject to} \quad 3x_1 + x_2 + R_1 = 3 \\
&\qquad\qquad\quad 4x_1 + 3x_2 - s_2 + R_2 = 6 \\
&\qquad\qquad\quad x_1 + 2x_2 + s_3 = 4 \\
&\qquad\qquad\quad x_1,\ x_2,\ s_2,\ s_3,\ R_1,\ R_2 \geq 0
\end{aligned}$$

2 The original Z row

Move the whole objective to one side, so every coefficient flips sign:

	x_1	x_2	s_2	s_3	R_1	R_2	Sol.
Z	−4	−1	0	0	−M	−M	0

The problem: R_1 and R_2 are basic, but its Z-row entry is not zero. It must become zero.

3 Fixing the Z row

It is a min problem, so multiply every R row by M and add it to the Z row. Add each column from top to bottom:

	x_1	x_2	s_2	s_3	R_1	R_2	Sol.
Old Z	−4	−1	0	0	−M	−M	0
+ M × R_1	3M	M	0	0	M	0	3M
+ M × R_2	4M	3M	−M	0	0	M	6M
New Z	7M − 4	4M − 1	−M	0	0	0	9M

Check: every R column is now 0 ✓

4 The first tableau

Basic	x_1	x_2	s_2	s_3	R_1	R_2	Sol.
R_1	3	1	0	0	1	0	3
R_2	4	3	−1	0	0	1	6
s_3	1	2	0	1	0	0	4
Z	$7M - 4$	$4M - 1$	$-M$	0	0	0	$9M$

5 Iteration 1

1. Entering: We look for the most positive entry in the Z row. Compare the coefficient of M first.

x_1: $7M - 4$ x_2: $4M - 1$

Entering: x_1

2. Leaving: Divide Sol by the entering column, positive entries only. The smallest ratio leaves.

Basic	x_1	Sol.	Ratio
R_1	3	3	1
R_2	4	6	$\frac{3}{2}$
s_3	1	4	4

Leaving: R_1 · The pivot = 3

3. The new pivot row

$$\text{New } x_1 = R_1 \div 3 = (1, \frac{1}{3}, 0, 0, \frac{1}{3}, 0 \mid 1)$$

4. The other rows: Each row = old row − (its entry in the entering column) × the new pivot row.

$$\text{New } R_2 = R_2 - (4) \times \text{New } x_1 = (0, \frac{5}{3}, -1, 0, -\frac{4}{3}, 1 \mid 2)$$

$$\text{New } s_3 = s_3 - (1) \times \text{New } x_1 = (0, \frac{5}{3}, 0, 1, -\frac{1}{3}, 0 \mid 3)$$

$$\text{New } Z = Z - (7M - 4) \times \text{New } x_1 = (0, \frac{5M}{3} + \frac{1}{3}, -M, 0, \frac{4}{3} - \frac{7M}{3}, 0 \mid 2M + 4)$$

5. The new tableau

Basic	x_1	x_2	s_2	s_3	R_1	R_2	Sol.
x_1	1	$\frac{1}{3}$	0	0	$\frac{1}{3}$	0	1
R_2	0	$\frac{5}{3}$	−1	0	$-\frac{4}{3}$	1	2
s_3	0	$\frac{5}{3}$	0	1	$-\frac{1}{3}$	0	3
Z	0	$\frac{5M}{3} + \frac{1}{3}$	−M	0	$\frac{4}{3} - \frac{7M}{3}$	0	2M + 4

6 Iteration 2

1. Entering: We look for the most positive entry in the Z row. Compare the coefficient of M first.

x_2: $\frac{5M}{3} + \frac{1}{3}$

Entering: x_2

2. Leaving: Divide Sol by the entering column, positive entries only. The smallest ratio leaves.

Basic	x_2	Sol.	Ratio
x_1	$\frac{1}{3}$	1	3
R_2	$\frac{5}{3}$	2	$\frac{6}{5}$
s_3	$\frac{5}{3}$	3	$\frac{9}{5}$

Leaving: R_2 · The pivot = $\frac{5}{3}$

3. The new pivot row

New $x_2 = R_2 \div \frac{5}{3} = (0,\ 1,\ -\frac{3}{5},\ 0,\ -\frac{4}{5},\ \frac{3}{5}\ |\ \frac{6}{5})$

4. The other rows: Each row = old row − (its entry in the entering column) × the new pivot row.

$$\text{New } x_1 = x_1 - \left(\frac{1}{3}\right) \times \text{New } x_2 = \left(1,\ 0,\ \frac{1}{5},\ 0,\ \frac{3}{5},\ -\frac{1}{5} \mid \frac{3}{5}\right)$$

$$\text{New } s_3 = s_3 - \left(\frac{5}{3}\right) \times \text{New } x_2 = (0,\ 0,\ 1,\ 1,\ 1,\ -1 \mid 1)$$

$$\text{New } Z = Z - \left(\frac{5M}{3} + \frac{1}{3}\right) \times \text{New } x_2 = \left(0,\ 0,\ \frac{1}{5},\ 0,\ \frac{8}{5} - M,\ -M - \frac{1}{5} \mid \frac{18}{5}\right)$$

5. The new tableau

Basic	x_1	x_2	s_2	s_3	R_1	R_2	Sol.
x_1	1	0	$\frac{1}{5}$	0	$\frac{3}{5}$	$-\frac{1}{5}$	$\frac{3}{5}$
x_2	0	1	$-\frac{3}{5}$	0	$-\frac{4}{5}$	$\frac{3}{5}$	$\frac{6}{5}$
s_3	0	0	1	1	1	−1	1
Z	0	0	$\frac{1}{5}$	0	$\frac{8}{5} - M$	$-M - \frac{1}{5}$	$\frac{18}{5}$

7 Iteration 3

1. Entering: We look for the most positive entry in the Z row.

s_2: $\frac{1}{5}$

Entering: s_2

2. Leaving: Divide Sol by the entering column, positive entries only. The smallest ratio leaves.

Basic	s_2	Sol.	Ratio
x_1	$\frac{1}{5}$	$\frac{3}{5}$	3
x_2	$-\frac{3}{5}$	$\frac{6}{5}$	— (not positive)
s_3	1	1	1

Leaving: s_3 · The pivot = 1

3. The new pivot row

$$\text{New } s_2 = s_3 \div 1 = (0,\ 0,\ 1,\ 1,\ 1,\ -1 \mid 1)$$

4. The other rows: Each row = old row − (its entry in the entering column) × the new pivot row.

$$\text{New } x_1 = x_1 - \left(\frac{1}{5}\right) \times \text{New } s_2 = \left(1,\ 0,\ 0,\ -\frac{1}{5},\ \frac{2}{5},\ 0 \mid \frac{2}{5}\right)$$

$$\text{New } x_2 = x_2 - \left(-\frac{3}{5}\right) \times \text{New } s_2 = \left(0,\ 1,\ 0,\ \frac{3}{5},\ -\frac{1}{5},\ 0 \mid \frac{9}{5}\right)$$

$$\text{New } Z = Z - \left(\frac{1}{5}\right) \times \text{New } s_2 = \left(0,\ 0,\ 0,\ -\frac{1}{5},\ \frac{7}{5} - M,\ -M \mid \frac{17}{5}\right)$$

5. The new tableau

Basic	x_1	x_2	s_2	s_3	R_1	R_2	Sol.
x_1	1	0	0	$-\frac{1}{5}$	$\frac{2}{5}$	0	$\frac{2}{5}$
x_2	0	1	0	$\frac{3}{5}$	$-\frac{1}{5}$	0	$\frac{9}{5}$
s_2	0	0	1	1	1	−1	1
Z	0	0	0	$-\frac{1}{5}$	$\frac{7}{5} - M$	−M	$\frac{17}{5}$

8 The answer

No positive entry is left in the Z row (M is large), so the tableau is optimal.

$$\text{Min } z = \frac{17}{5} \quad \text{at} \quad x_1 = \frac{2}{5}, \quad x_2 = \frac{9}{5}$$

05 Solution 5: max with ≤ and ≥

1 Standard form + R

`≤` takes only `$s_1$`, and `≥` takes `$-s_2$` and `R`. In max, R gets `−M`.

$$\begin{aligned} &\text{Maximize } z = 3x_1 + 2x_2 + 0s_1 + 0s_2 - MR \\ &\text{subject to} \quad x_1 + x_2 + s_1 = 4 \\ &\qquad\qquad\quad x_1 + 3x_2 - s_2 + R = 6 \\ &\qquad\qquad\quad x_1, x_2, s_1, s_2, R \geq 0 \end{aligned}$$

2 The original Z row

Move the whole objective to one side, so every coefficient flips sign:

	x_1	x_2	s_1	s_2	R	Sol.
Z	−3	−2	0	0	M	0

The problem: `R` is basic, but its Z-row entry is not zero. It must become zero.

3 Fixing the Z row

It is a max problem, so multiply every R row by M and subtract it from the Z row.

Work each column from top to bottom:

	x_1	x_2	s_1	s_2	R	Sol.
Old Z	−3	−2	0	0	M	0
− M × R	−M	−3M	0	M	−M	−6M
New Z	−M − 3	−3M − 2	0	M	0	−6M

Check: every R column is now 0 ✓

4 The first tableau

Basic	x_1	x_2	s_1	s_2	R	Sol.
s_1	1	1	1	0	0	4
R	1	3	0	−1	1	6
Z	−M − 3	−3M − 2	0	M	0	−6M

5 Iteration 1

1. Entering: We look for the most negative entry in the Z row. Compare the coefficient of M first.

x_1: $-M - 3$ x_2: $-3M - 2$

Entering: x_2

2. Leaving: Divide Sol by the entering column, positive entries only. The smallest ratio leaves.

Basic	x_2	Sol.	Ratio
s_1	1	4	4
R	3	6	2

Leaving: R · The pivot = 3

3. The new pivot row

$$\text{New } x_2 = R \div 3 = \left(\frac{1}{3}, 1, 0, -\frac{1}{3}, \frac{1}{3} \mid 2\right)$$

4. The other rows: Each row = old row − (its entry in the entering column) × the new pivot row.

$$\text{New } s_1 = s_1 - (1) \times \text{New } x_2 = \left(\frac{2}{3}, 0, 1, \frac{1}{3}, -\frac{1}{3} \mid 2\right)$$

$$\text{New } Z = Z - (-3M - 2) \times \text{New } x_2 = \left(-\frac{7}{3}, 0, 0, -\frac{2}{3}, M + \frac{2}{3} \mid 4\right)$$

5. The new tableau

Basic	x_1	x_2	s_1	s_2	R	Sol.
s_1	$\frac{2}{3}$	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	2
x_2	$\frac{1}{3}$	1	0	$-\frac{1}{3}$	$\frac{1}{3}$	2
Z	$-\frac{7}{3}$	0	0	$-\frac{2}{3}$	$M + \frac{2}{3}$	4

6 Iteration 2

1. Entering: We look for the most negative entry in the Z row.

$x_1: -\frac{7}{3}$ $s_2: -\frac{2}{3}$

Entering: x_1

2. Leaving: Divide Sol by the entering column, positive entries only. The smallest ratio leaves.

Basic	x_1	Sol.	Ratio
s_1	$\frac{2}{3}$	2	3
x_2	$\frac{1}{3}$	2	6

Leaving: s_1 · The pivot = $\frac{2}{3}$

3. The new pivot row

$$\text{New } x_1 = s_1 \div \frac{2}{3} = (1,\ 0,\ \frac{3}{2},\ \frac{1}{2},\ -\frac{1}{2} \mid 3)$$

4. The other rows: Each row = old row − (its entry in the entering column) × the new pivot row.

$$\text{New } x_2 = x_2 - (\frac{1}{3}) \times \text{New } x_1 = (0,\ 1,\ -\frac{1}{2},\ -\frac{1}{2},\ \frac{1}{2} \mid 1)$$

$$\text{New } Z = Z - (-\frac{7}{3}) \times \text{New } x_1 = (0,\ 0,\ \frac{7}{2},\ \frac{1}{2},\ M - \frac{1}{2} \mid 11)$$

5. The new tableau

Basic	x_1	x_2	s_1	s_2	R	Sol.
x_1	1	0	$\frac{3}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	3
x_2	0	1	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	1
Z	0	0	$\frac{7}{2}$	$\frac{1}{2}$	$M - \frac{1}{2}$	11

7 The answer

No negative entry is left in the Z row (M is large), so the tableau is optimal.

$$\text{Max } z = 11 \quad \text{at} \quad x_1 = 3, \quad x_2 = 1$$

06 Solution 6: no feasible solution

1 Standard form + R

$\leq$ takes only s_1, and $\geq$ takes $-s_2$ and R. In max, R gets $-M$.

$$\begin{aligned} &\text{Maximize } z = 2x_1 + x_2 + 0s_1 + 0s_2 - MR \\ &\text{subject to} \quad x_1 + x_2 + s_1 = 2 \\ &\qquad\qquad\quad x_1 + x_2 - s_2 + R = 4 \\ &\qquad\qquad\quad x_1, x_2, s_1, s_2, R \geq 0 \end{aligned}$$

2 The original Z row

Move the whole objective to one side, so every coefficient flips sign:

	x_1	x_2	s_1	s_2	R	Sol.
Z	−2	−1	0	0	M	0

The problem: `R` is basic, but its Z-row entry is not zero. It must become zero.

3 Fixing the Z row

It is a max problem, so multiply every R row by M and subtract it from the Z row.

Work each column from top to bottom:

	x_1	x_2	s_1	s_2	R	Sol.
Old Z	−2	−1	0	0	M	0
− M × R	−M	−M	0	M	−M	−4M
New Z	−M − 2	−M − 1	0	M	0	−4M

Check: every R column is now 0 ✓

4 The first tableau

Basic	x_1	x_2	s_1	s_2	R	Sol.
s_1	1	1	1	0	0	2
R	1	1	0	−1	1	4
Z	−M − 2	−M − 1	0	M	0	−4M

5 Iteration 1

1. Entering: We look for the most negative entry in the Z row. Compare the coefficient of M first.

```
x₁: −M − 2      x₂: −M − 1
```

Entering: `x₁`

2. Leaving: Divide Sol by the entering column, positive entries only. The smallest ratio leaves.

Basic	x_1	Sol.	Ratio
s_1	1	2	2
R	1	4	4

Leaving: `s₁` · The pivot = `1`

3. The new pivot row

```
New x₁ = s₁ ÷ 1 = (1, 1, 1, 0, 0 | 2)
```

4. The other rows: Each row = old row − (its entry in the entering column) × the new pivot row.

```
New R = R − (1) × New x₁ = (0, 0, −1, −1, 1 | 2)
```

```
New Z = Z − (−M − 2) × New x₁ = (0, 1, M + 2, M, 0 | 4 − 2M)
```

5. The new tableau

Basic	x_1	x_2	s_1	s_2	R	Sol.
x_1	1	1	1	0	0	2
R	0	0	−1	−1	1	2
Z	0	1	M + 2	M	0	4 − 2M

6 The answer

No negative entry is left in the Z row (M is large), so the tableau is optimal.

The answer

But `R` is still basic with value `2`, not zero. So the problem has **no feasible solution**.